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comes in the form of system-in-package (SIP). While system-on-module (SoM) designs can provide a functional solution, these are generally PCB-based and have to be mounted as a daughter card with a specific connector for all input-output. Unfortunately, this increases the cost and size of the design, neither of which is desirable.

SIP solutions have all the power and functionality of SoM solutions, but these are packaged in normal, monolithic IC packaging and can be handled in the same way, including the ability to be soldered to a PCB as part of an automated manufacturing process.

Octavo Systems' OSD335x series of SIPs are perfectly capable of acting as the core of a powerful and sophisticated system, while being as easy to implement as an MCU. At their core is a Texas Instruments Sitara SoC that comprises a 1GHz ARM Cortex-A8 processor, two programmable real-time units and DDR3 memory. There is also a power management IC (PMIC)

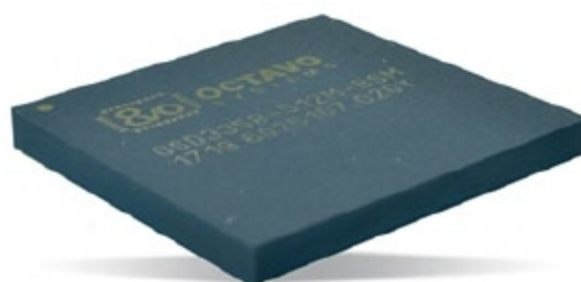


Fig. 2: Octavo Systems' OSD335x

to control the complex power needs. It includes multiple power rails that need to be accurately sequenced while ensuring efficient operation. Engineers simply need to connect a single power feed as they would do with a traditional MCU.

OSD335x PMIC is capable of multiplexing multiple power sources—a useful feature if the product can be powered from different power sources such as an AC adaptor and a battery. It can also manage the battery, saving the need for the designers to create this functionality themselves.

By fully integrating the design into a single SIP, there are other benefits that might not be available in a discrete or SoM approach. For example,

miniature packaging and a high level of integration reduce the energy used to transfer data between the core and the memory.

In our modern world, products need to offer greater functionality and often integrate directly into the Internet of Things (IoT) to be successful. SIP solutions such as Octavio OSD335x have far greater power and functionality than traditional MCUs, but these are just as easy to use. The Octavio family offers multiple variants that have been carefully tailored to meet a wide variety of applications and design challenges, and these are all compact enough to fit the small spaces available in many IoT applications.

IoT-ready design tools including Beaglebone Black Wireless have Octavio OSD335x. Hence, once the design is completed using the platform, the SIP can easily be implemented on the final production PCB with relatively little effort. The inbuilt Sitara SoC has significant software support, reducing the task of coding and getting designs to market faster. **EFY**